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Atomic layer self aligned substrate processing and integrated toolset

Abstract

Apparatus and methods to process one or more wafers are described. A substrate is exposed to a plurality of process stations to deposit, anneal, treat and optionally etch a film in small increments to provide self-aligned growth of the film on a substrate surface.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. application Ser. No. 16/412,696, filed May 15, 2019, which claims priority to U.S. Provisional Application No. 62/672,560, filed May 16, 2018, the entire disclosures of which are hereby incorporated by reference herein.

TECHNICAL FIELD

(1) The present disclosure relates generally to apparatus for depositing thin films. In particular, the disclosure relates to apparatus having a plurality of separate processing stations to deposit a self-aligned film on a substrate.

BACKGROUND

(2) Current atomic layer deposition (ALD) processes have a number of potential issues and difficulties. Many ALD chemistries (e.g., precursors and reactants) are “incompatible”, which means that the chemistries cannot be mixed together. If the incompatible chemistries mix, a chemical vapor deposition (CVD) process, instead of the ALD process could occur. The CVD process generally has less thickness control than the ALD process and/or can result in the creation of gas phase particles which can cause defects in the resultant device. For a traditional time-domain ALD process in which a single reactive gas is flowed into the processing chamber at a time, a long purge/pump out time occurs so that the chemistries are not mixed in the gas phase. A spatial ALD chamber can move one or more wafer(s) from one environment to a second environment faster than a time-domain ALD chamber can pump/purge, resulting in higher throughput.

(3) With electronic device scaling (e.g., <10 nm), it is extremely hard to form self-aligned features. Any misalignment results in shorting, ruining the device performance. Additionally, self-aligned processes, such as silicide, etc. result in lateral growth due to large diffusion. The lateral growth can also result in shorting. Current state of the art self-aligned schemes use multiple processes, such as deposition, anneal, removal, to create self-aligned features.

(4) Therefore, there is a need in the art for improved deposition apparatus and methods of forming self-aligned films with little or no misalignment of films.

SUMMARY

(5) One or more embodiments of the disclosure are directed to processing tools comprising a plurality of process stations. Each process station provides a processing region separated from processing regions of adjacent process stations. A substrate support has a support surface to support a wafer for processing. The substrate support is configured to move the wafer between at least two of the plurality of process stations. A controller is connected to the substrate support and the plurality of process stations. The controller is configured to activate the substrate support to move the wafer between stations, and to control a process occurring in each of the process stations. The plurality of process stations comprises a deposition station, an anneal station, and a treatment station.

(6) Additional embodiments of the disclosure are directed to methods for depositing a film. A substrate is moved to a deposition station to deposit a film on a surface of the substrate. The substrate is moved to an anneal station to anneal the film on the substrate. The substrate is moved to a treatment station to treat the annealed film with a plasma. Each of the deposition station, anneal station and treatment station are part of an integrated processing tool with a controller configured to move the substrate, deposit the film, anneal the film and treat the annealed film.

(7) Further embodiments of the disclosure are directed to methods for depositing a film. A substrate having a first substrate surface and a second substrate surface is provided in a deposition station. The first substrate surface comprises a different material than the second substrate surface. A film is deposited on the first substrate surface and the second substrate surface in the deposition station. The film has a thickness less than or equal to about 20 Å. The substrate is moved from the deposition station to an anneal station to anneal the film and form an annealed film. The substrate is moved to a treatment station to treat the annealed film with a plasma to form a treated annealed

film. The plasma changes at least one property of the film on at least one of the first substrate surface or the second substrate surface. The substrate is moved to an etch station to selectively etch the film from the second substrate surface relative to the first substrate surface. Depositing the film, annealing the film, treating the film and selectively etching the film are repeated to selectively deposit a film having a thickness greater than or equal to about 1000 Å on the first substrate surface.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of this disclosure and are therefore not to be considered limiting of its scope, for the disclosure may admit to other equally effective embodiments.

(2) FIG. 1A shows a schematic representation of a processing tool in accordance with one or more embodiment of the disclosure;

(3) FIGS. 1B through 1H illustrate a deposition process in accordance with one or more embodiment of the disclosure;

(4) FIG. 1J illustrates a flowchart of the deposition process illustrated in FIGS. 1B through 1H in accordance with one or more embodiment of the disclosure;

(5) FIG. 2 shows a bottom perspective view of a support assembly in accordance with one or more embodiment of the disclosure;

(6) FIG. 3 shows a top perspective view of a support assembly in accordance with one or more embodiment of the disclosure;

(7) FIG. 4 shows a top perspective view of a support assembly in accordance with one or more embodiment of the disclosure;

(8) FIG. 5 shows a schematic cross-sectional view of the support assembly of FIG. 4 taken along line IV-IV;

(9) FIG. 6 shows a cross-sectional perspective view of a processing chamber in accordance with one or more embodiment of the disclosure;

(10) FIG. 7 shows a cross-sectional view of a processing chamber in accordance with one or more embodiment of the disclosure;

(11) FIG. 8 shows a schematic representation of a processing platform in accordance with one or more embodiment of the disclosure;

(12) FIGS. 9A through 9I shows a schematic views of process stations in a processing chamber in accordance with one or more embodiment of the disclosure; and

(13) FIGS. 10A and 10B shows a schematic representation of process in accordance with one or more embodiment of the disclosure.

DETAILED DESCRIPTION

(14) Before describing several exemplary embodiments of the disclosure, it is to be understood that the disclosure is not limited to the details of construction or process steps set forth in the following description. The disclosure is capable of other embodiments and of being practiced or being carried out in various ways.

(15) A “substrate” as used herein, refers to any substrate or material surface formed on a substrate upon which film processing is performed during a fabrication process. For example, a substrate surface on which processing can be performed include materials such as silicon, silicon oxide, strained silicon, silicon on insulator (SOI), carbon doped silicon oxides, amorphous silicon, doped

silicon, germanium, gallium arsenide, glass, sapphire, and any other materials such as metals, metal nitrides, metal alloys, and other conductive materials, depending on the application. Substrates include, without limitation, semiconductor wafers. Substrates may be exposed to a pretreatment process to polish, etch, reduce, oxidize, hydroxylate, anneal and/or bake the substrate surface. In addition to film processing directly on the surface of the substrate itself, in the present disclosure, any of the film processing steps disclosed may also be performed on an under-layer formed on the substrate as disclosed in more detail below, and the term “substrate surface” is intended to include such under-layer as the context indicates. Thus for example, where a film/layer or partial film/layer has been deposited onto a substrate surface, the exposed surface of the newly deposited film/layer becomes the substrate surface.

(16) As used in this specification and the appended claims, the terms “precursor”, “reactant”, “reactive gas” and the like are used interchangeably to refer to any gaseous species that can react with the substrate surface, or with a film formed on the substrate surface.

(17) Some embodiments of the disclosure provide integrated toolsets which allow for the fabrication of self-aligned features based on underlying substrate materials. Some embodiments allow for the growth of different films on different features or surfaces (e.g., metal silicide on metal and SiN on dielectric). In some embodiments, the integrated tool comprises multiple stations with or without a rotating platform to deposit, anneal, treat the surface and optional removal processes. The sequence can be repeated to allow for very controlled growth in the feature without creating lateral growth (encroachment). Embodiments of the disclosure can be used with planar substrates, substrates with features (e.g., vias, trenches, fins) and with hardmask/patterning applications. A planar application might form a metal silicide film on a metal surface and a nitride film on an adjacent dielectric surface. An application using surface features include, but are not limited to, formation of vias over metal/oxide surfaces so that a metal silicide forms on the metal and a nitride forms on the oxide. In an exemplary hardmask/patterning application in which a metal is on a spacer material, a metal silicide can be formed on the bottom and top surfaces.

(18) FIG. 1A illustrates an integrated processing tool **10** for forming self-aligned features. The processing tool **10** has a plurality of process stations **11**, **12**, **13**, **14** with each station providing a processing region **11a**, **12a**, **13a**, **14a** separated from adjacent process stations. The exemplary embodiment illustrated has four stations; however, the skilled artisan will recognize that there can be more or less than four stations. The individual stations can be separated from adjacent stations by gas curtains or physical barriers.

(19) A substrate support **15** (shown as a dashed line) has a support surface to support a substrate or wafer for processing. The substrate support is configured to move a wafer between at least two of the plurality of processing stations. In some embodiments, the substrate support is configured to move the wafer between all of the process stations. As used in this manner, the term “between” includes the processing regions of the individual process stations.

(20) A controller **16** can be connected to the substrate support **15** and the plurality of process stations **11**, **12**, **13**, **14**. The controller can be configured to activate the substrate support **15** to move the wafer between stations, and to control a process occurring in each of the process stations. In some embodiments, the plurality of process stations **11**, **12**, **13**, **14** include, respectively, a deposition station, an annealing station, a treatment station and an optional etch station.

(21) Referring to FIGS. 1B through 1H, an exemplary process is illustrated with a planar substrate having two different surface chemistries. FIG. 1J illustrates a flowchart of the process **500** illustrated in FIGS. 1B through 1H. At **510**, the substrate is provided, or positioned, in an environment for processing. For example, the substrate can be positioned in the process station **11** and is therefore provided for processing. As shown in FIG. 1B, the substrate **21** has a first material **22** with a first surface **22a** and a second material **23** with a second surface **23a** that is different than the first material **22** and the first surface **22a**. The process station **11** can include any suitable deposition chamber that can form the film. In some embodiments, the deposition station comprises

one or more of an atomic layer deposition (ALD) chamber, plasma enhanced atomic layer deposition (PEALD), a chemical vapor deposition (CVD) chamber, or a plasma enhanced chemical vapor deposition (PECVD) chamber. In some embodiments, the first material **22** comprises a metal (e.g., cobalt, copper, titanium). In some embodiments, the second material **23** comprises a dielectric (e.g., an oxide).

(22) In some embodiments, the process stations may comprise exposure to a portion of a deposition process. In some embodiments, process station **11** may expose a substrate to a first reactant and process station **12** may expose the substrate to a second reactant to react with the first reactant and deposit a film. In this regard, two or more stations may be used for a single deposition process.

(23) At **520**, in the deposition chamber of process station **11**, a film **24** is formed on the substrate **21**, as shown in FIG. **1C**. The film **24** can be formed conformally so that there is a substantially equal thickness on both the first material **22** and the second material **23**, or can be selective to the first material **22** relative to the second material **23**. The degree of selectivity can be in the range of about 1:1 to about 50:1 for the first material **22**: second material **23**.

(24) The film **24** can be formed to any suitable thickness. In some embodiments, the film **24** has a thickness less than or equal to about one monolayer of the material being deposited. In some embodiments, the thickness of the film **24** is greater than 0.1 Å up to about 10 Å, 15 Å, 20 Å, 25 Å, 30 Å, 35 Å or 40 Å. In some embodiments, the film comprises one or more of silicon, titanium, copper, cobalt, tungsten or aluminum.

(25) After formation of the film **24**, the substrate **21** is moved from process station **11** to process station **12**. As shown in FIG. **1D** and at **530**, the film **24** can be exposed to an anneal process in process station **12** to form an annealed film **25**. In some embodiments, the anneal station comprises one or more of a laser anneal, thermal anneal or flash anneal chamber.

(26) After forming the annealed film **25**, the substrate **21** is moved from process station **12** to process station **13**. As shown in FIG. **1E** and at **540**, the annealed film **25** is treated to form treated film **26**. The treatment can be any suitable treatment depending, for example, the film composition. In some embodiments, the treatment comprises a plasma processing chamber. The plasma changes at least one property of the annealed film **25**. In some embodiments, the treatment changes a property of the annealed film **25** on the first surface **22a** differently than on the second surface **23a** so that there are differences between the treated film **26a** and treated film **26b**.

(27) In some embodiments, as shown in FIG. **1F**, the treatment removes the annealed film from the second surface **23a**. In these embodiments, the substrate **21** may be processed without an etch process (described below). In an embodiment of this sort, the process can repeat by moving the substrate back to the process station **11**.

(28) In some embodiments, the processing tool **10** includes an etch station as process station **14**. In an embodiment like that of FIG. **1E** in which the properties of the film are different on the first surface **22a** than on the second surface **23a**, the substrate **21** can be moved from the process station **13** to process station **14**. As shown in FIG. **1G** and at **550**, in some embodiments, the substrate **21** is exposed to an etch process which can selectively remove the treated film **26b** from the second surface **23a** relative to the treated film **26a** from the first surface **22a**. As illustrated in FIG. **1G**, the thickness of the treated film **26a** may be reduced as part of the etch process, while the treated film **26b** is substantially completely removed (>95% by weight).

(29) The etch station can be any suitable etch chamber that can selectively remove the film **26b** from the second surface **23a** relative to the film **26a** from the first surface **22a**. In some embodiments, the etch station comprises one or more of a chemical etch, a reactive ion etch or an isotropic etch chamber.

(30) At **560**, it is determined whether a predetermined thickness of film **26a** has been formed. If not, the process **500** returns to **520** to deposit film **24** on the substrate. If a predetermined thickness has been formed, as shown in FIG. **1H**, the process **500** continues to **570** for optional further processing.

(31) In some embodiments, the thickness of the film **26a** is measured through an inline or external process. In some embodiments, the thickness of the film **26a** is measured in situ. In some embodiments, the thickness of the film **26a** is determined by measuring one or more of vertical thickness, critical dimension (CD), spacer width and/or spacer height. In some embodiments, the predetermined thickness of film **26a** is formed through a number of repeated cycles.

(32) In some embodiments, the deposition, anneal, treatment and optional etch processes can be repeated to form a film **26a** of a predetermined thickness, as shown in FIG. **1H**. The predetermined thickness of some embodiments is greater than or equal to about 100 Å, 200 Å, 300 Å, 400 Å, 500 Å, 600 Å, 700 Å, 800 Å, 900 Å or 1000 Å.

(33) In some embodiments, the controller **16** includes a central processing unit, a memory, and support circuits. The controller **16** may control the process stations or processing chambers directly, or via computers (or controllers) associated with particular process chamber and/or support system components. The controller **16** may be one of any form of general-purpose computer processor that can be used in an industrial setting for controlling various chambers and sub-processors. The memory or computer readable medium of the controller may be one or more of readily available memory such as random access memory (RAM), read only memory (ROM), floppy disk, hard disk, optical storage media (e.g., compact disc or digital video disc), flash drive, or any other form of digital storage, local or remote. The support circuits are coupled to the CPU for supporting the processor in a conventional manner. These circuits include cache, power supplies, clock circuits, input/output circuitry and subsystems, and the like. One or more processes may be stored in the memory as software routine that may be executed or invoked to control the operation of the apparatus or individual components in the manner described herein. The controller **16** can include one or more configurations which can include any commands or functions to control flow rates, gas valves, gas sources or other processes for performing the various configurations. The various configurations of the controller can allow control of the process stations and movements of the substrate support through one or more motors, actuators, valves, flow controllers and/or heaters to enable the controller to execute the configuration.

(34) In some embodiments, the controller **16** has one or more configurations to operate the processing tool **10** including process stations **11**, **12**, **13**, **14** and substrate support **15**. In some embodiments, the controller comprises one or more of: a first configuration to move a wafer sequentially from the deposition chamber to the anneal chamber to the treatment chamber; a second configuration to deposit a layer on a substrate in the deposition chamber; a third configuration to anneal a layer on the substrate in the anneal chamber; and a fourth configuration to plasma treat the annealed layer in the treatment chamber. In some embodiments, the process stations include an etch station and the first configuration moves a wafer sequentially from the deposition chamber to the anneal chamber to the treatment chamber to the etch chamber, and the controller has a fifth configuration to perform an etch process on a wafer in the etch chamber.

(35) In some embodiments, the controller is configured to deposit a film in the deposition chamber to a thickness (e.g., less than or equal to about 15 Å), move the substrate to the anneal chamber to anneal the film, move the substrate to the treatment chamber to expose the film to plasma, and move the substrate to the etch chamber to selectively etch the film from some portions of the substrate.

(36) In some embodiments, the controller is configured to repeat deposition, anneal, treatment and etch processes to build a film of a predetermined thickness.

(37) One or more embodiments of the disclosure use spatial separation between two or more processing environments. Some embodiments advantageously provide apparatus and methods to maintain separation of incompatible gases. Some embodiments advantageously provide apparatus and methods including optimizable plasma processing. Some embodiments advantageously provide apparatus and methods that allow for a differentiated thermal dosing environment, a differentiated plasma treatment environment and other environments.

(38) One or more embodiments of the disclosure are directed to processing chambers having four processing environments. Some embodiments have more than four and some embodiments have less than four. The processing environments can be mounted coplanar to the wafer(s) that are moving in a horizontal plane. The process environments are placed in a circular arrangement. A rotatable structure with one to four (or more) individual wafer heaters mounted thereon moves the wafers in a circular path with a diameter similar to the process environments. Each heater may be temperature controlled and may have one or multiple concentric zones. For wafer loading, the rotatable structure could be lowered so that a vacuum robot could pick finished wafers and place unprocessed wafers on pins located above each wafer heater (in the lower Z position). In operation, each wafer can be under an independent environment until the process is finished, then rotatable structure can rotate to move the wafers on the heaters to the next environment (90° rotation for four stations, 120° rotation if three stations) for processing.

(39) Some embodiments of the disclosure advantageously provide spatial separation for ALD with incompatible gases. Some embodiments allow for higher throughput and tool resource utilization than a traditional time-domain or spatial process chamber. Each process environment can operate at a different pressure. The heater rotation has Z direction motion so each heater can be sealed into a chamber.

(40) Some embodiments advantageously provide plasma environments that can include one or more of microwave, ICP, parallel plate CCP or 3 electrode CCP. The entire wafer can be immersed in plasma; eliminating the plasma damage from non-uniform plasma across the wafer.

(41) In some embodiments, a small gap between the showerhead and the wafer can be used to increase dose gas utilization and cycle time speed. Precise showerhead temperature control and high operating range (up to 230° C.). Without being bound by theory, it is believed that the closer the showerhead temperature is to the wafer temperature, the better the wafer temperature uniformity.

(42) The showerheads can include small gas holes (<200 μm), a high number of gas holes (many thousands to greater than 10 million) and recursively fed gas distribution inside the showerhead using small distribution volume to increase speed. The small size and high number gas holes can be created by laser drilling or dry etching. When a wafer is close to the showerhead, there is turbulence experienced from the gas going through the vertical holes towards the wafer. Some embodiments allow for a slower velocity gas through the showerhead using a large number of holes spaced close together achieving a uniform distribution to the wafer surface.

(43) Some embodiments are directed to integrated processing platforms using a plurality of chambers on a single tool. The processing platform can have a variety of chambers that can perform different processes.

(44) Some embodiments of the disclosure are directed to apparatus and methods to move wafer(s) attached to a wafer heater(s) from one environment to another environment. The rapid movement can be enabled by electrostatically chucking (or clamping) the wafer(s) to the heater(s). The movement of the wafers can be in linear or circular motion.

(45) Some embodiments of the disclosure are directed to methods of processing one or more substrates. Examples include, but are not limited to, running one wafer on one heater to a plurality of different sequential environments spatially separated; running two wafers on two wafer heaters to three environments (two environments the same and one different environment between the two similar environments); wafer one sees environment A then B, and repeats, while wafer two sees B then A and repeats; one environment remaining idle (without wafer); running two wafers in two first environments and two second environments where both wafers see the same environments at the same time (i.e., both wafers in A then both go to B); four wafers with two A and two B environments; and two wafers processing in A's while the other two wafers are processing in B's. In some embodiments, wafers are exposed to environment A and environment B repeatedly, and then exposed to a third environment located in the same chamber.

(46) In some embodiments, wafers go through a plurality of chambers for processing where at least one of the chambers does sequential processing with a plurality of spatially separated environments within the same chamber.

(47) Some embodiments are directed to apparatus with spatially separated processing environments within the same chamber where the environments are at significantly different pressures (e.g., one at <100 mT another at >3T). In some embodiments, the heater rotation robot moves in the z-axis to seal each wafer/heater into the spatially separated environments.

(48) Some embodiments include a structure built above the chamber with a vertical structural member applying a force upward to the center of the chamber lid to eliminate deflection caused by the pressure of atmosphere on the topside and the vacuum on the other side. The magnitude of force of the structure above can be mechanically adjusted based on the deflection of the top plate. The force adjustment can be done automatically using a feedback circuit and force transducer or manually using, for example, a screw that can be turned by an operator.

(49) FIGS. 2 through 6 illustrate support assemblies **100** in accordance with one or more embodiments of the disclosure. The support assembly **100** includes a rotatable center base **110**. The rotatable center base **110** can have a symmetrical or asymmetrical shape and defines a rotational axis **111**. The rotational axis **111**, as can be seen in FIG. 5, extends in a first direction. The first direction may be referred to as the vertical direction; however, it will be understood that the use of the term “vertical” in this manner is not limited to a direction normal to the pull of gravity.

(50) The support assembly **100** includes at least two support arms **120** connected to and extending from the center base **110**. The support arms **120** have an inner end **121** and an outer end **122**. The inner end **121** is in contact with the center base **110** so that when the center base **110** rotates around the rotational axis **111**, the support arms **120** rotate as well. The support arms **120** can be connected to the center base **110** at the inner end **121** by fasteners (e.g., bolts) or by being integrally formed with the center base **110**.

(51) In some embodiments, the support arms **120** extend orthogonal to the rotational axis **111** so that one of the inner ends **121** or outer ends **122** are further from the rotational axis **111** than the other of the inner ends **121** and outer ends **122** on the same support arm **120**. In some embodiments, the inner end **121** of the support arm **120** is closer to the rotational axis **111** than the outer end **122** of the same support arm **120**.

(52) The number of support arms **120** in the support assembly **100** can vary. In some embodiments, there are at least two support arms **120**. In some embodiments, there are three support arms **120**. In some embodiments, there are four support arms **120**. In some embodiments, there are five support arms **120**. In some embodiments, there are six support arms **120**.

(53) The support arms **120** can be arranged symmetrically around the center base **110**. For example, in a support assembly **100** with four support arms **120**, each of the support arms **120** are positioned at 90° intervals around the center base **110**. In a support assembly **100** with three support arms **120**, the support arms **120** are positioned at 120° intervals around the center base **110**.

(54) A heater **130** is positioned at the outer end **122** of the support arms **120**. In some embodiments, each support arm **120** has a heater **130**. The center of the heaters **130** are located at a distance from the rotational axis **111** so that upon rotation of the center base **110** the heaters **130** move in a circular path.

(55) The heaters **130** have a support surface **131** which can support a wafer. In some embodiments, the heater **130** support surfaces **131** are substantially coplanar. As used in this manner, the term “substantially coplanar” means that the planes formed by the individual support surfaces **131** are within ±5°, ±4°, ±3°, ±2° or ±1° of the planes formed by the other support surfaces **131**.

(56) In some embodiments, the heaters **130** are positioned directly on the outer end **122** of the support arms **120**. In some embodiments, as illustrated in the drawings, the heaters **130** are elevated above the outer end **122** of the support arms **120** by a heater standoff **134**. The heater standoffs **134** can be any size and length to increase the height of the heaters **130**.

(57) In some embodiments, a channel **136** is formed in one or more of the center base **110**, the support arms **120** and/or the heater standoffs **134**. The channel **136** can be used to route electrical connections or to provide a gas flow.

(58) The heaters can be any suitable type of heater known to the skilled artisan. In some embodiments, the heater is a resistive heater with one or more heating elements within a heater body.

(59) The heaters **130** of some embodiments include additional components. For example, the heaters may comprise an electrostatic chuck. The electrostatic chuck can include various wires and electrodes so that a wafer positioned on the heater support surface **131** can be held in place while the heater is moved. This allows a wafer to be chucked onto a heater at the beginning of a process and remain in that same position on that same heater while moving to different process regions.

(60) The heater **130** and support surface **131** can include one or more gas outlets to provide a flow of backside gas. This may assist in the removal of the wafer from the support surface **131**. As shown in FIGS. 2 and 3, the support surface **131** includes a plurality of openings **137** and a gas channel **138**. The openings **137** and/or gas channel **138** can be in fluid communication with one or more of a vacuum source or a gas source (e.g., a purge gas).

(61) Some embodiments of the support assembly **100** include a sealing platform **140**. The sealing platform has a top surface **141**, a bottom surface and a thickness. The sealing platform **140** can be positioned around the heaters **130** to help provide a seal or barrier to minimize gas flowing to a region below the support assembly **100**. In some embodiments, as shown in FIG. 3, the sealing platforms **140** are ring shaped and are positioned around each heater **130**. In the illustrated embodiment, the sealing platforms **140** are located below the heater **130** so that the top surface **141** of the sealing platform **140** is below the support surface **131** of the heater. In some embodiments, as shown in FIGS. 4 and 5, the sealing platform **140** is a single component that surrounds all of the heaters **130** with a plurality of openings **142** to allow access to the support surface **131** of the heaters **130**. The openings **142** can allow the heaters to pass through the sealing platform **140**. In some embodiments, the sealing platform **140** is fixed so that the sealing platform **140** moves vertically and rotates with the heaters **130**. In some embodiments, the sealing platform **140** has a top surface **141** forming a major plane that is substantially parallel with a major plane formed by the support surface **131** of the heater **130**, as shown in FIG. 5. In some embodiments, the sealing platform **140** has a top surface **141** forming a major plane that is a distance above the major plane of the support surface **131** by an amount substantially equal to the thickness of a wafer to be processed so that the wafer surface is coplanar with the top surface **141** of the sealing platform **140**, as shown in FIG. 4.

(62) In some embodiments, as shown in FIGS. 4 and 5, the sealing platform **140** is supported by support post **127**. The support post **127** may have utility in preventing sagging of the center of the sealing platform **140** when a single component platform is used.

(63) In some embodiments, as illustrated in FIG. 7, the support assembly **100** includes at least one motor **150**. The at least one motor **150** is connected to the center base **110** and is configured to rotate the support assembly **100** around the rotational axis **111**. In some embodiments, the at least one motor is configured to move the center base **110** in a direction along the rotational axis **111**. For example, in FIG. 7, motor **155** is connected to motor **150** and can move the support assembly **100** in the Z-axis or vertically.

(64) Referring to FIGS. 6 and 7, one or more embodiments of the disclosure are directed to processing chambers **200** that incorporate the support assembly **100**. The processing chamber **200** has a housing **202** with walls **204**, a bottom **206** and a top **208** which define an interior volume **209**. The embodiment illustrated in FIG. 6 does not show the top **208**.

(65) The processing chamber **200** includes a plurality of process stations **210**. The process stations **210** are located in the interior volume **209** of the housing **202** and are positioned in a circular arrangement around the rotational axis **111**. Each process station **210** comprises a gas injector **212**

having a front face **214**. In some embodiments, the front faces **214** of each of the gas injectors **212** are substantially coplanar.

(66) The process stations **210** can be configured to perform any suitable process and provide any suitable process conditions. The type of gas injector **212** used will depend on, for example, the type of process being performed and the type of process chamber. For example, a process station **210** configured to operate as an atomic layer deposition apparatus may have a showerhead or vortex type injector. Whereas, a process station **210** configured to operate as a plasma station may have one or more electrode and grounded plate configuration to generate a plasma while allowing a plasma gas to flow toward the wafer. The embodiment illustrated in FIG. 7 has a different type of process station **210** on the left side of the drawing than on the right side of the drawing. Suitable process stations **210** include, but are not limited to, thermal processing stations, microwave plasma, three-electrode CCP, ICP, parallel plate CCP, UV exposure, laser processing, pumping chambers, annealing stations and metrology stations.

(67) As shown in FIG. 7, one or more vacuum streams and purge gas streams can be used to help isolate one process station **210** from an adjacent process station **210**. A purge gas plenum **260** is in fluid communication with a purge gas port **261** at the outer boundary of the process stations **210**. A vacuum plenum **265** is in fluid communication with a vacuum port **266**. The purge gas port **261** and the vacuum port **266** can extend around the perimeter of the process station **210** to form a gas curtain. The gas curtain can help minimize or eliminate leakage of process gases into the interior volume **209**.

(68) The number of process stations **210** can vary with the number of heaters **130** and support arms **120**. In some embodiments, there are an equal number of heaters **130**, support arms **120** and process stations **210**. In some embodiments, the heaters **130**, support arms **120** and process stations **210** are configured to that each of the support surfaces **131** of the heaters **130** can be located adjacent the front faces **214** of different process stations **210** at the same time. Stated differently, each of the heaters is positioned in front of a process station at the same time.

(69) FIG. 8 shows a processing platform **300** in accordance with one or more embodiment of the disclosure. The embodiment shown in FIG. 8 is merely representative of one possible configuration and should not be taken as limiting the scope of the disclosure. For example, in some embodiments, the processing platform **300** has different numbers of process chambers **200**, buffer stations **320** and robot **330** configurations.

(70) The exemplary processing platform **300** includes a central transfer station **310** which has a plurality of sides **311**, **312**, **313**, **314**. The transfer station **310** shown has a first side **311**, a second side **312**, a third side **313** and a fourth side **314**. Although four sides are shown, those skilled in the art will understand that there can be any suitable number of sides to the transfer station **310** depending on, for example, the overall configuration of the processing platform **300**.

(71) The transfer station **310** has a robot **330** positioned therein. The robot **330** can be any suitable robot capable of moving a wafer during processing. In some embodiments, the robot **330** has a first arm **331** and a second arm **332**. The first arm **331** and second arm **332** can be moved independently of the other arm. The first arm **331** and second arm **332** can move in the x-y plane and/or along the z-axis. In some embodiments, the robot **330** includes a third arm or a fourth arm (not shown). Each of the arms can move independently of other arms.

(72) The embodiment illustrated includes six processing chamber **200** with two each connected to the second side **312**, third side **313** and fourth side **314** of the central transfer station **310**. Each of the processing chambers **200** can be configured to perform different processes.

(73) The processing platform **300** can also include one or more buffer station **320** connected to the first side **311** of the central transfer station **310**. The buffer stations **320** can perform the same or different functions. For example, the buffer stations may hold a cassette of wafers which are processed and returned to the original cassette, or one of the buffer stations may hold unprocessed wafers which are moved to the other buffer station after processing. In some embodiments, one or

more of the buffer stations are configured to pre-treat, pre-heat or clean the wafers before and/or after processing.

(74) The processing platform **300** may also include one or more slit valves **318** between the central transfer station **310** and any of the processing chambers **200**. The slit valves **318** can open and close to isolate the environment within the processing chamber **200** from the environment within the central transfer station **310**. For example, if the processing chamber will generate plasma during processing, it may be helpful to close the slit valve for that processing chamber to prevent stray plasma from damaging the robot in the transfer station.

(75) The processing platform **300** can be connected to a factory interface **350** to allow wafers or cassettes of wafers to be loaded into the processing platform **300**. A robot **355** within the factory interface **350** can be used to move the wafers or cassettes into and out of the buffer stations. The wafers or cassettes can be moved within the processing platform **300** by the robot **330** in the central transfer station **310**. In some embodiments, the factory interface **350** is a transfer station of another cluster tool (i.e., another multiple chamber processing platform).

(76) A controller **395** may be provided and coupled to various components of the processing platform **300** to control the operation thereof. The controller **395** can be a single controller that controls the entire processing platform **300**, or multiple controllers that control individual portions of the processing platform **300**. For example, the processing platform **300** may include separate controllers for each of the individual processing chambers **200**, central transfer station **310**, factory interface **350** and robots **330**. In some embodiments, the controller **395** includes a central processing unit (CPU) **396**, a memory **397**, and support circuits **398**. The controller **395** may control the processing platform **300** directly, or via computers (or controllers) associated with particular process chamber and/or support system components. The controller **395** may be one of any form of general-purpose computer processor that can be used in an industrial setting for controlling various chambers and sub-processors. The memory **397** or computer readable medium of the controller **395** may be one or more of readily available memory such as random access memory (RAM), read only memory (ROM), floppy disk, hard disk, optical storage media (e.g., compact disc or digital video disc), flash drive, or any other form of digital storage, local or remote. The support circuits **398** are coupled to the CPU **396** for supporting the processor in a conventional manner. These circuits include cache, power supplies, clock circuits, input/output circuitry and subsystems, and the like. One or more processes may be stored in the memory **398** as software routine that may be executed or invoked to control the operation of the processing platform **300** or individual processing chambers in the manner described herein. The software routine may also be stored and/or executed by a second CPU (not shown) that is remotely located from the hardware being controlled by the CPU **396**.

(77) FIGS. **9A** through **9I** illustrate various configurations of processing chambers **200** with different process stations **210**. The lettered circles represent the different process stations **210** and process conditions. For example, in FIG. **9A**, there are four process stations **210** each with a different letter. This represents four process stations **210** with each station having different conditions than the other stations. As indicated by the arrow, a process could occur by moving the heaters with wafers from stations A through D. After exposure to D, the cycle can continue or reverse.

(78) In FIG. **9B**, four wafers can be processed at the same time with the wafers being moved on the heaters back and forth between the A and B positions. Two wafers could start in the A positions and two wafers in the B positions. The independent process stations **210** allow for the two of the stations to be turned off during the first cycle so that each wafer starts with an A exposure.

(79) The embodiment illustrated in FIG. **9B** might also be useful in processing two wafers in the four process stations **210**. This might be particularly useful if one of the processes is at a very different pressure or the A and B process times are very different.

(80) In FIG. **9C**, three wafers might be processed in a single processing chamber **200** in and ABC

process. One station can either be turned off or perform a different function (e.g., pre-heating). (81) In FIG. 9D, two wafers can be processed in an AB-Treat process. For example, wafers might be placed on the B heaters only. A quarter turn clockwise will place one wafer in the A station and the second wafer in the T station. Turning back will move both wafers to the B stations and another quarter turn counter-clockwise will place the second wafer in the A station and the first wafer in the B station.

(82) In FIG. 9E, up to four wafers can be processed at the same time. For example, if the A station is configured to perform a CVD or ALD process, four wafers can be processed simultaneously.

(83) FIGS. 9F through 9I show similar types of configurations for a processing chamber 200 with three process stations 210. Briefly, in FIG. 9F, a single wafer (or more than one) can be subjected to an ABC process. In FIG. 9G, two wafers can be subjected to an AB process by placing one in the A position and the other in one of the B positions. The wafers can then be moved back and forth so that the wafer starting in the B position moves to the A position in the first move and then back to the same B position. In FIG. 9H a wafer can be subjected to an AB-Treat process. In FIG. 9I, three wafers can be processed at the same time.

(84) FIGS. 10A and 10B illustrate another embodiment of the disclosure. In FIG. 10A, the heater 130 on support arm 120 has been rotated to a position beneath process station 210 so that wafer 101 is adjacent the gas injector 212. An O-ring 129 on the support arm 120, or on an outer portion of the heater 130, is in a relaxed state. The support arm 120 and heater 130 are moved toward the process station 210 so that the support surface 131 of the heater 130 is moved to contact or nearly contact the front face 214 of the process station 210, as shown in FIG. 10B. In this position, O-ring 129 is compressed forming a seal around the outer edge of the support arm 120 or outer portion of the heater 130. This allows the wafer 101 to be moved as close the injector 212 as possible to minimize the volume of the reaction region 219 so that the reaction region 219 can be rapidly purged.

(85) Gases which might flow out of the reaction region 219 are evacuated through vacuum port 266 into vacuum plenum 265 and to an exhaust or foreline. A purge gas curtain outside of the vacuum port 266 can be generated by purge gas plenum 260 and purge gas port 261. Additionally, a purge gas can be flowed through gap 237 between the heater 130 and the support arm 120 to further curtain off the reaction region 219 and prevent reactive gases from flowing into the interior volume 209 of the processing chamber 200.

(86) Reference throughout this specification to “one embodiment,” “certain embodiments,” “one or more embodiments” or “an embodiment” means that a particular feature, structure, material, or characteristic described in connection with the embodiment is included in at least one embodiment of the disclosure. Thus, the appearances of the phrases such as “in one or more embodiments,” “in certain embodiments,” “in one embodiment” or “in an embodiment” in various places throughout this specification are not necessarily referring to the same embodiment of the disclosure.

Furthermore, the particular features, structures, materials, or characteristics may be combined in any suitable manner in one or more embodiments.

(87) Although the disclosure herein has been described with reference to particular embodiments, it is to be understood that these embodiments are merely illustrative of the principles and applications of the present disclosure. It will be apparent to those skilled in the art that various modifications and variations can be made to the method and apparatus of the present disclosure without departing from the spirit and scope of the disclosure. Thus, it is intended that the present disclosure include modifications and variations that are within the scope of the appended claims and their equivalents.

Claims

1. A method for depositing a film, the method comprising: loading a substrate onto a substrate support of an integrated processing tool, the integrated processing tool comprising a deposition

station, an anneal station, a treatment station, an etching station, and a controller, and the substrate support comprising a rotatable center defining a rotation axis, at least two support arms extending from the center base, each of the support arms having an inner end in contact with the center base and an outer end; and a heater positioned on the outer end of each of the support arms, the heaters having a support surface; moving the substrate to the deposition station; depositing a film on a surface of the substrate in the deposition station; moving the substrate to the anneal station; annealing the film on the substrate in the anneal station to form an anneal film; moving the substrate to the treatment station; treating the annealed film with a plasma in the treatment station to form a treated annealed film; moving the substrate to the etching station; and etching the treated annealed film to at least partially remove the film from the surface of the substrate wherein the controller activates the substrate support to move the substrate, deposit the film, anneal the film, treat the annealed film, and etch the treated annealed film.

2. The method of claim 1, wherein a cycle of depositing the film, annealing the film, treating the annealed film and etching the treated annealed film is repeated a plurality of times to selectively form the film on a first surface of the substrate relative to a second surface of the substrate.

3. The method of claim 2, wherein the film is deposited to a thickness less than or equal to about 20 Å and the cycle is repeated greater than or equal to about 20 times.

4. A method for depositing a film, the method comprising: loading a substrate onto a substrate support of an integrated processing tool, the integrated processing tool comprising a deposition station, an anneal station, a treatment station, an etching station, and a controller, and the substrate support comprising a rotatable center defining a rotation axis, at least two support arms extending from the center base, each of the support arms having an inner end in contact with the center base and an outer end; and a heater positioned on the outer end of each of the support arms, the heaters having a support surface, and the substrate having a first substrate surface and a second substrate surface, the first substrate surface comprising a different material than the second substrate surface; moving the substrate to the deposition station; depositing a film on the first substrate surface and the second substrate surface in the deposition station, the film having a thickness less than or equal to about 20 Å; moving the substrate from the deposition station to an anneal station; annealing the film to form an annealed film; moving the substrate to a treatment station; treating the annealed film with a plasma to form a treated annealed film, the plasma changing at least one property of the film on at least one of the first substrate surface or the second substrate surface; moving the substrate to an etch station; selectively etching the treated annealed film from the second substrate surface relative to the first substrate surface; and repeating depositing the film, annealing the film, treating the film, selectively etching the film, and corresponding movements to selectively deposit the film with a thickness greater than or equal to about 1000 Å on the first substrate surface.
